Transport Policy 25 (2013) 138–147

Contents lists available at SciVerse ScienceDirect

Transport Policy

journal homepage: www.elsevier.com/locate/tranpol

Subjective valuation of the transit transfer experience: The case of Santiago de Chile

Francisca Javiera Navarrete*, Juan de Dios Ortúzar

Pontificia Universidad Católica de Chile, Department of Transport Engineering and Logistics, Casilla 306, Código 105, Santiago 22, Santiago, Chile

ARTICLE INFO

Available online 16 January 2013

Keywords:
Public transport transfer
Stated choice
Efficient design
Mixed logit
Transantiago

ABSTRACT

The still controversial *Transantiago* public transport system, in Santiago de Chile, has a topological structure that often requires its users to make one or more transfers to reach their destinations. This was rarely necessary in the previous non-integrated system and users reacted with unexpected displeasure when it started even though fare integration in the new system means that transferring involved no extra costs. This study investigates users' subjective valuations of the transfer experience and its associated elements (walking and waiting times), analysing how these vary for different types of transfer combinations. In particular, we determine the relative preferences the following transfer combinations: metro–metro, metro–bus, bus–metro and bus–bus, with emphasis on the importance of various physical characteristics such as information availability, the existence of station escalators and the ability to board the first available bus or train. We also estimate the relative valuation of the different time component values (walking, waiting and in-vehicle) of trips including a transfer, and also derived the penalties users assign to trips that require transferring at intermodal stations during the morning peak hour. The trip time components most heavily penalised were the walking time involved in transferring and the final walking time to the user's destination.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

Transfers between different public transport modes or lines are a significant structural element in integrated urban transit systems, the topological designs of which typically combine a trunk network of high-frequency and high-capacity services, with a secondary feeder network of lower frequency and lower capacity services. In such systems, transferring is often necessary when travelling across different zones. The advantage of this design is that it optimises resources while reducing congestion and pollution, but it comes at the price of forcing many users to change modes or lines at some point on their journeys.

Despite the obvious importance of the transfer phenomenon not many studies appear to have addressed it in the past (CTPS, 1997; Liu et al., 1997; Wardman et al., 2001). More recently, a thesis by Crockett (2002) analysed the willingness of transit users to transfer as a function of different service levels, network information, security and waiting times in a non-integrated system comprising an urban rail network and buses in the city of Chicago. Guo (2003) added user-assigned penalties to transfers based on travel and walking times within the rail network serving central Boston. Espino et al. (2007) identified and analysed users' preferences over various transfer alternatives for urban and interurban travel on non-fare-integrated bus lines and examined their willingness-to-pay (WTP) for a direct trip. Finally, Guo (2008) studied transit systems planning and its effect on modal split in Boston's interurban/commuter rail network and the London Underground. No further work was found despite a thorough review of the literature.

The principal objective of this paper is to present a critical analysis of the various elements of transferring that matter to passengers on peak-hour journeys. Using data from *Transantiago*, the radically new and quite controversial public transport system serving Santiago de Chile (Muñoz et al., 2009), we set out to determine the most relevant variables in the transfer experience and their relative weights, and the associated user time valuations (walking, waiting, in-vehicle travel), placing particular emphasis on the physical characteristics that users transferring must face. An essential element of our work was the creation of a data bank from information gathered through a stated choice (SC) survey. This was constructed using an efficient design methodology under a Bayesian parameter estimation framework (Rose and Bliemer, 2009).

The remainder of this paper is organised as follows. Section 2 describes the theoretical background of our study; Section 3 describes the experiments conducted, the design and application

* Corresponding author. Tel.: +56 2 354 4270.
E-mail addresses: fjnavarr@puc.cl (F.J. Navarrete), jos@ing.puc.cl (J. de D. Ortúzar).

0967-070X/$ - see front matter © 2012 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.tranpol.2012.10.006

of the SC survey, the proposed models and the analysis of the results they generated; finally, Section 4 presents our main conclusions.

2. Methodological background

2.1. Discrete choice modelling

Random utility theory postulates that individuals (q) choose among different alternatives (A_j) on the basis of their utility U_{jq}. The modeller, an observer of the system, knows only some of the elements considered by the individual (indeed, this means that some individual choices often cannot be explained) and is forced to assume that the individual's utility has two elements: first, a representative utility function V_{jq}, which is considered linear in its parameters in the simplest version of the theory (i.e. assuming compensatory behaviour):

$$V_{jq} = \sum_k \theta_{jk} X_{jkq} \tag{1}$$

where θ are the parameters (marginal utilities) to be estimated and $\boldsymbol{X}$ are attributes of the alternatives or socioeconomic characteristics of the individuals as observed by the modeller. The second element is a random error term ε_{jq} such that:

$$U_{jq} = V_{jq} + \varepsilon_{jq} = \sum_k \theta_{jk} X_{jkq} + \varepsilon_{jq} \tag{2}$$

If the errors distribute identically and independently (iid) Extreme Value Type 1, the probability that individuals q choose alternative A_i from their available choice sets $A_{(q)}$ is given by the multinomial logit (MNL) model (see Domencich and McFadden, 1975):

$$P_{iq} = \frac{\exp(\beta V_{iq})}{\sum_{A_j \in A_{(q)}} \exp(\beta V_{jq})} \tag{3}$$

where β is a non-identifiable scale factor that has to be normalised. This model is readily estimated using the maximum likelihood method, which yields the set of parameters that is most likely to replicate the observed sample (Ortúzar and Willumsen, 2011). Unfortunately, the MNL model has significant limitations, such as assuming that the alternatives are mutually independent (the independence of irrelevant alternatives property), that the utility functions of the alternatives are homoscedastic, that all sampled individuals have the same tastes (i.e. equal θ_{ik}) and that all observations are independent. If any of these restrictions is not satisfied, the model may produce biased results.

The mixed logit (ML) model, on the other hand, is capable of a much more general representation because it can be extended to handle situations with correlation between alternatives, heteroscedasticity, taste variations and repeated observations of individuals (i.e. panel effect). Two specifications of the ML model have been posited: the random parameters logit (RPL) and the error components logit (ECL). The first has a very similar structure to an MNL model except that the parameters θ, instead of being fixed for all individuals, are distributed over the population. This characteristic allows the differences in individual tastes for the various utility function attributes to be easily modelled. The RPL is given by the following structure (Train, 2009):

$$U_{jq} = \sum_k \overline{\theta}_{jk} X_{jkq} + \sum_k \gamma_{qk} X_{jkq} + \varepsilon_{jq} \tag{4}$$

where $\overline{\theta}_{jk}$ represents the population mean for the kth attribute and γ_{qk} its standard deviation. These two are used to construct a parameter θ_{jkq} that represents the individual marginal utility for attribute k of alternative j.

The error components (ECL) specification, on the other hand, consists basically in adding an error term η_{jq} to the traditional random utility formulation in Eq. (2) that can be distributed as the modeller sees fit. This model is structured very generally as follows:

$$U_{jq} = \sum_k \theta_{jk} X_{jkq} + \varepsilon_{jq} + \eta_{jq} \tag{5}$$

However, estimating any type of ML model is considerably more complex than estimating a MNL given that the former's probability function is not a closed analytic expression such as Eq. (3) but rather the integral of a probability function (MNL or some other function if so desired) over the range of variation of the parameters. To solve the multivariable integral and find the model's best estimators, the simulated maximum likelihood method is most commonly employed (Godoy and Ortúzar, 2008), but the model can also be estimated using a Bayesian approach (Train, 2009).

2.2. Willingness-to-pay estimation

A variety of methods are available to estimate the WTP for improving an attribute X_{jk} and their complexities are many, but all have in common that they require an attribute related to the price or direct cost of the good or service in question.

For MNL and ECL models with linear utilities such as Eq. (1), the WTP formula is

$$WTP(X_{jk}) = \frac{\theta_{jk}}{\theta_{jc}} \tag{6}$$

where θ_{jk} is the parameter of the attribute for which WTP is being estimated and θ_{jc} is the cost parameter (Jara-Díaz, 2007). But the values in Eq. (6) are based on point parameter estimates which, even in the relatively simple MNL case, have an asymptotically Normal distribution (as they are maximum likelihood estimators). To circumvent this problem Armstrong et al. (2001) proposed the following simple and highly efficient method of finding the WTP confidence interval:

$$\Delta WTP(X_{jk}) = \left(\frac{\theta_{jk}}{\theta_{jc}}\frac{t_c}{t_k}\right)\left(\frac{(t_c t_k - \rho t_\alpha{}^2)}{(t_c{}^2 - t_\alpha{}^2)} \pm \frac{\sqrt{(\rho t_\alpha{}^2 - t_c t_k)^2 - (t_k{}^2 - t_\alpha{}^2)(t_c{}^2 - t_\alpha{}^2)}}{(t_c{}^2 - t_\alpha{}^2)}\right) \tag{7}$$

where t_k and t_c are the respective t-test values for θ_{jk} and θ_{jc}, t_α is the tabulated critical value for a confidence level α and ρ is the correlation coefficient between the two parameters. The estimation of WTP for RPL models is much more complex. The interested reader may wish to consider the discussion in Sillano and Ortúzar (2005).

2.3. Stated choice survey

Stated choice (SC) surveys are a powerful modelling tool, particularly in cases where the individuals surveyed cannot be confronted with the real-life situation under study. In this type of survey individuals can be queried on a number of different choice situations, yielding various responses per respondent, with the consequent savings in costs compared to revealed preference (RP) surveys which normally produce only a single answer per respondent. However, with SC data in MNL models we would be forced to assume that the responses of each individual to different choice situations are independent, and for many years this problem was ignored in the literature (see Ortúzar et al., 2000). A relatively simple solution to this problem is to estimate ML models using an error component to correlate the responses of each respondent.

To obtain quality data from an SC survey, the experimental design must be well-adapted to the situation under study as it presents the respondent with hypothetical alternatives. The attributes of these are allowed to vary within a set range, which is realistic yet sufficiently wide to provide unbiased parameter estimates. The design must specify *a priori* the number of such alternatives in each choice situation, how many choice situations will be included and how many individuals will be surveyed (Caussade et al., 2005; Rose et al., 2009b).

In recent years, the use of efficient design techniques has come to predominate (Rose and Bliemer, 2009). Unlike orthogonal designs, which are appropriate for linear models, efficient designs minimize some standard error measurement for the estimated parameters. These designs lead to better estimators for a given number of respondents but require *a priori* values for each parameter (or at least an indication of the expected sign) to "guide the design" towards the optimum. Since these values heavily influence the final outcome, a Bayesian approach is recommended in which relatively imprecise initial values are used to construct one or more pilot surveys, generating estimates that will be employed in the definitive version (Torres et al., in press).

3. Application to Transantiago

The current transit network in Santiago de Chile went into operation during the midst of the summer school holidays on 10 February 2007. Known as *Transantiago*, its implementation brought profound changes to the existing public transport system (Muñoz et al., 2009). Generating problems as well as benefits, it posed a major challenge to user comprehension of the new setup, a fundamental requisite in a city of approximately 6 million inhabitants.

The topological design of the new system altered the traditional way of using public transport in Santiago, in that most journeys now required making at least one transfer[1]. It triggered profound changes in both the management of the network and the travel habits of the transit-using public who were previously accustomed to making long trips with no transfers (a practice closely linked to the high operating costs associated with the length of the former routes, which could stretch as much as 30 km between outlying points on opposite sides of the city; see Muñoz et al. (2009)).

Although *Transantiago* has many advantages over the old "yellow buses" thanks to fare integration and improved accessibility in many areas of the city, users have still not fully adapted to the new way of travelling. The main stumbling block is the need to transfer, which is perceived negatively due to its consequences in terms of congestion, uncertainty and waiting times (Muñoz et al., 2009). The transfer experience is therefore essential to detecting and understanding the new individual perceptions of the transit system as it currently operates.

3.1. Design and application of our stated choice survey

Our SC experiment consisted in showing only single transfer alternatives to respondents. Preliminary analyses revealed that if direct trips were also offered people would always choose them even if total travel time was higher. Multiple transfers were ruled out due to their complexity.

The variables used in the design may be divided into two groups: (i) quantitative variables, which took values within an *a priori*-defined finite set, and (ii) qualitative (binary) variables, which could take a value of 0 or 1 depending on whether or not a given characteristic was present. The quantitative variables considered were the following:

- *Cost*: Fare (Ch$) charged to each user to make a complete trip. The fare varies a little during the day (i.e. there are peak and off-peak fares) and if the trip includes a Metro stage the fare is approximately 10% dearer. Finally, the fare allows making three transfers but only one of those by Metro (a Metro–Metro transfer does not count as a transfer in this case).
- *Initial Walk Time* (IAT): time (min) elapsed from the moment a person sets out from home to the moment they reach the stop or platform to await the first transport mode vehicle/carriage.
- *Initial Wait Time* (IWT): time (min) elapsed from the moment a person reaches the stop or platform to the moment they board the first mode.
- *Initial Travel Time* (ITT): time (min) elapsed from the moment a person boards the first mode to the moment they alight from it.
- *Transfer Walk Time* (TAT): time (min) elapsed from the moment a person alights the first mode to the moment they reach the stop or platform to await the second mode.
- *Transfer Wait Time* (TWT): time (min) elapsed from the moment a person reaches the stop or platform of the second mode to the moment they are able to board it.
- *Final Travel Time* (FTT): time (min) elapsed from the moment a person boards the second mode to the moment they alight from it.
- *Final Walk Time* (FAT): time (min) elapsed from the moment a person alights the second mode until they arrive at their final destination (place of work of study, etc.)

The qualitative variables considered were the following:

- *Intermodal Station*: takes the value 1 if the transfer was made at an intermodal transfer station and 0 if it was made at a normal stop or station[2].
- *Escalator Available*: takes the value 1 if there are escalators or elevators for changing levels at transfer stations and 0 otherwise; for bus–bus transfers at surface-level stops (whether ordinary or in a "paid zone"[3]), the variable is always 0.
- *First Available Bus or Train* (or simply "*First Available*"): takes the value 1 if a passenger can board the first bus or metro wagon that stops (that is, there is room on the bus or train for the passenger to board) and 0 otherwise.
- *Information Available*: takes the value 1 if static or dynamic travel information is available on routes and transfers and 0 otherwise.

The basic structure of the operation modelled through the quantitative and qualitative variables is shown in Fig. 1. Our goal was to capture the impacts of the above-described variables in

[1] A transfer can be thought of as either a mode change, or a line change within the same mode, in both cases on a single trip. The concept embraces not only the fact of alighting one transit vehicle and boarding another but also the time invested in the change, the distance involved and the waiting time for the new vehicle.

[2] Although in Santiago there are only a few proper intermodal stations, in the survey we also considered as such stations where the transfer is undoubtedly better than in a normal bus stop or just Metro transfer station.

[3] Fare "paid zones" are bus stops at which passengers pay their fare before boarding the bus, that is, upon entering a defined area at the bus stop. This permits faster boarding when the bus arrives as all doors can be used for entering the vehicle, allowing more passengers to board simultaneously.

order to evaluate their influences on transferring between public-transport modes.

The possible values for the quantitative variables are laid out in Table 1. To make sure that the situations presented to the survey respondents were realistic, the times and costs used were familiar values to them, that is, within their actual experience. With this in mind we generated four predefined survey forms corresponding to four different total travel times (20, 40, 60 and 80 min) that together covered a range of trip typologies representing the majority of system users. Once a respondent indicated their trip duration the appropriate survey could be administered by the interviewer. Survey generation was carried out using *Ngene* (Rose et al., 2009a).

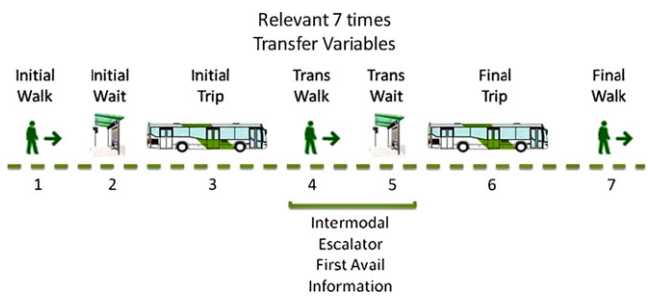

Fig. 1. Graphical representation of the basic variables modelled.

Due to the high number of variables in the model, i.e., 12 including *Cost* plus an alternative specific constant (ASC) for each of the transfer combinations Metro–Metro, Metro–bus and bus–Metro in the efficient design (bus–bus was left as reference option), each respondent would have needed to answer 16 choice situations to guarantee good results. However, earlier studies have found that in the Chilean context a survey of such length would place too heavy a cognitive load on respondents (Caussade et al., 2005; Rose and Bliemer, 2009). As a result, our design contained two blocks of eight choice situations each (Louviere et al., 2000), and since there were four alternative trip durations, we ended up with eight different questionnaires.

Before carrying out the final survey we conducted focus group interviews and two pilot versions (Navarrete, 2010). The final survey was applied face to face using Excel (see Fig. 2) at

Table 1
Possible values of quantitative variables by type of survey (depending in approximated total trip length).

Variable	20 min survey	40 min survey	60 min survey	80 min survey
Cost	400;460;520;580	400;460;520;580	400;460;520;580	400;460;520;580
Initial Walk	1;2;3;4	2;3;4;5	2;4;6;8	3;5;7;9
Initial Wait	1;2;3;4	3;4;5;6	3;5;7;9	3;5;7;9
Initial Trip	4;5;6;7	8;10;12;14	14;16;18;20	20;24;28;32
Trans Walk	1;2;3;4	2;3;4;5	2;4;6;8	3;5;7;9
Trans Wait	1;2;3;4	3;4;5;6	3;5;7;9	3;5;7;9
Final Trip	4;5;6;7	8;10;12;14	14;16;18;20	20;24;28;32
Final Walk	1;2;3;4	2;3;4;5	2;4;6;8	3;5;7;9

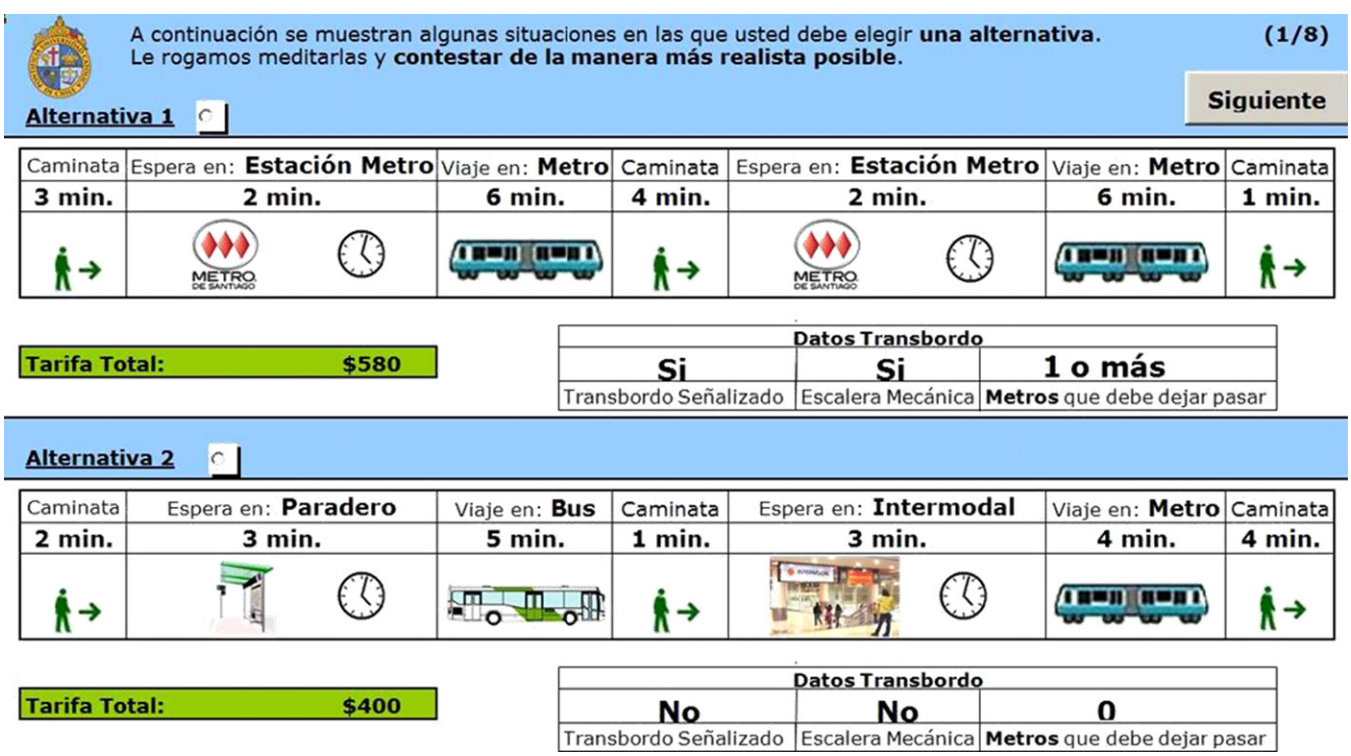

Fig. 2. Survey page: choice of travel alternative.

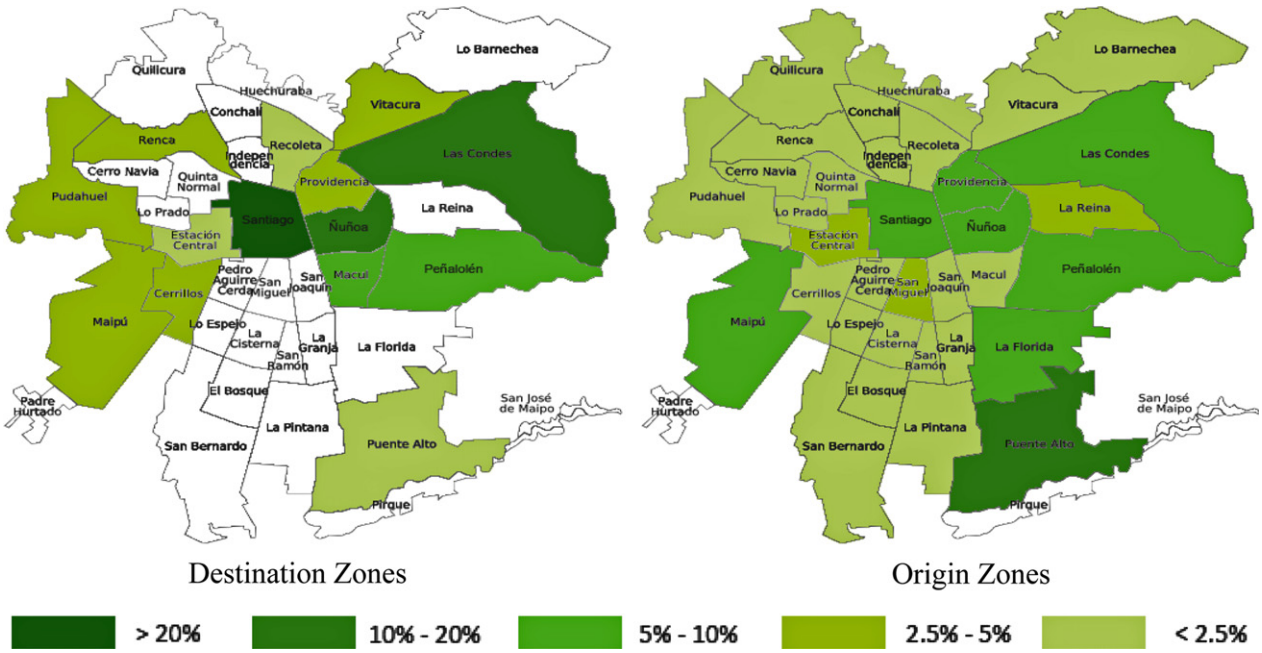

Fig. 3. Zones of Santiago: destination and origin of respondents.

workplaces[4] in 14 urban districts across Santiago, in an attempt to include as wide a variety of social backgrounds as possible and observe the variations in the perceptions of transit service provision at different points around the city.

Although respondent destinations were located in just 14 of the city's 37 administrative districts, they originated from 34 of these (see Fig. 3). This diversity is a key element in securing perceptions from users facing a range of different services levels in *Transantiago* (which unfortunately still vary from zone to zone), thereby reducing bias and ensuring more representative parameter estimates.

The final survey was administered during the months of July and August 2009 to a sample of 240 individuals. We detected 26 lexicographic respondents (i.e. either total time or fare minimisers), leaving 214[5] respondents with truly compensatory choices. This provided a database of 1712 observations, each individual responding to eight choice situations.

3.2. Structure of the discrete choice models

All our models were estimated using BIOGEME (Bierlaire, 2003). The first one was a basic MNL formulation (MNL1) with the following linear utility:

$$\begin{aligned} V_i = {} & \theta_{MetroMetro} \cdot D_{MM} + \theta_{MetroBus} \cdot D_{MB} + \theta_{BusMetro} \cdot D_{BM} \\ & + \theta_{Cost} \cdot Cost + \theta_{InitialWalk} \cdot Initial\ Walk \\ & + \theta_{InitialWait} \cdot Initial\ Wait + \theta_{InitialTrip} \cdot Initial\ Trip \\ & + \theta_{TransWalk} \cdot Trans\ Walk + \theta_{TransWait} \cdot Trans\ Wait \\ & + \theta_{FinalTrip} \cdot Final\ Trip + \theta_{FinalWalk} \cdot Final\ Walk \\ & + \theta_{Intermodal} \cdot Intermodal + \theta_{Escalator} \cdot Escalator \\ & + \theta_{FirstAvail} \cdot First\ Avail + \theta_{Info} \cdot Information \end{aligned} \tag{8}$$

We then proceeded to estimate a second MNL model with systematic taste variations (i.e. interactions between the experimental attributes and the respondents' socioeconomic variables). The structure of this model is not linear in its parameters, as is the case with MNL1; rather, its behaviour is given by the following equation (Ortúzar and Willumsen, 2011):

$$V_{jq} = \sum_k \left(\theta_{jk} + \sum_{n=1}^{N} \theta_{k_SEn} SE_{nq} \right) X_{jkq} \tag{9}$$

where SE_{nq} is the value of socioeconomic variable n for individual q. The specification of the SE variables included in the model is:

- *Carlic*: ratio of the number of cars to the number of licensed drivers in the individual's household with a ceiling of 1.
- *Age*: is equal to 1 for individuals between the ages of 18 and 29 and 0 otherwise.
- *Gender*: takes the value of 1 for women and 0 otherwise.
- *Employment*: takes the value 1 if the individual is an employee and 0 if self-employed.
- *N°Trans*: Number of transfers that respondents declare to make in their regular trip to work.
- *Wage Rate*: Variable reflecting directly the individual's value of work time (Jara-Díaz and Ortúzar, 1989). It was calculated as follows:

$$w = \frac{Net\ \ income[Ch\$/month]}{4.333[weeks/month] \cdot Hours\ \ worked\ \ per\ \ week \cdot 60[\mathrm{min}/hour]}$$

Three further variables were defined to represent the proportions of walking, waiting and in-vehicle (travel) time in the total trip time. Their values can range from 0 to 1.

$$Walk\ Prop_{Ind} = \frac{Total\ Walk_{Ind}}{Total\ Walk_{Ind} + Total\ Wait_{Ind} + Total\ In{-}Vehicle_{Ind}} \tag{10}$$

$$Wait\ Prop_{Ind} = \frac{Total\ Wait_{Ind}}{Total\ Walk_{Ind} + Total\ Wait_{Ind} + Total\ In{-}Vehicle_{Ind}} \tag{11}$$

[4] Workplace interviews have a long tradition in Chile and their effectiveness has been demonstrated in a series of research studies at our Department (see, among others, Ortúzar and Hutt, 1984; Iragüen and Ortúzar, 2004).

[5] The efficient design used required a minimum of 199 respondents to guarantee that the critical variable (that with the greatest variance) would attain the necessary level of significance; in our case that variable was *Information Available*.

$$In{-}Vehicle\ Prop_{Ind} = \frac{Total\ In{-}Vehicle_{Ind}}{Total\ Walk_{Ind} + Total\ Wait_{Ind} + Total\ In{-}Vehicle_{Ind}} \tag{12}$$

These variables serve to capture the weight attached to each trip segment for individuals making trips of the same total duration but with differing proportions of walking, waiting and in-vehicle travel time.

A considerable number of specifications were estimated before one was chosen to be model MNL2 (see Appendix). Its formulation, given in Eq. (13), accommodates 14 interactions between basic model attributes and several of the individuals' socioeconomic variables:

$$\begin{aligned} V_i = {} & \theta_{MM} \cdot D_{MetroMetro} + \theta_{MB} \cdot D_{MetroBus} + \theta_{BM} \cdot D_{BusMetro} \\ & + (\theta_{Cost} + \theta_{Cost_Age} \cdot Age_{<30}) \cdot Cost + (\theta_{Initial\ Walk} \\ & + \theta_{IWalk_CarLic} \cdot CarLic) \cdot Initial\ Walk + (\theta_{Initial\ Wait} \\ & + \theta_{IWait_CarLic} \cdot CarLic) \cdot Initial\ Walk + \theta_{Inicial\ Trip} \cdot Inicial\ Trip \\ & + (\theta_{TransWalk} + \theta_{TWalk_Gender} \cdot Gender_{Female}) \cdot Trans\ Walk \\ & + \theta_{Trans\ Wait} \cdot Trans\ Wait - \theta_{FinalTrip} \cdot Final\ Trip \\ & + (\theta_{Final\ Walk} + \theta_{FWalk_Gender} \cdot Gender_{Female}) \cdot Final\ Walk \\ & + (\theta_{Info} + \theta_{Inf_Age} \cdot Age_{<30}) \cdot Information \\ & + (\theta_{Escalator} + \theta_{Esc_Gender} \cdot Gender_{Female} \\ & + \theta_{Esc_N^{\circ}Trans} \cdot N^{\circ}Trans) \cdot Escalator \\ & + \begin{pmatrix} \theta_{\text{First Avail}} + \theta_{\text{FAvail_CarLic}} \cdot CarLic + \\ \theta_{\text{FAvail_WaitProp}} \cdot Wait\ Prop \end{pmatrix} \cdot First\ Avail \\ & + \begin{pmatrix} \theta_{\text{Intermodal}} + \theta_{\text{Intermodal_Gender}} \cdot Gender_{Female} + \\ \theta_{\text{Intermodal_Ocup}} \cdot Employment_{Staff} + \theta_{\text{Intermodal_WageRate}} \cdot Wage\ Rate \\ + \theta_{\text{Intermodal_WalkProp}} \cdot Walk\ Prop \end{pmatrix} \\ & \times Intermodal \end{aligned} \tag{13}$$

This model was built from the base model, MNL1, by incorporating systematic taste variations for most of the defined attributes. Since all variables were found to have the expected signs as well as reasonable magnitudes, MNL2 allows adding important information on individual segmentation. Although many variables had low *t*-test values, we left them in the specification since they help to explain social phenomena that were captured at the focus groups and pilot surveys. The statistical details of models MNL1 and MNL2 are given in the Appendix; a likelihood ratio (LR) test (Ortúzar and Willumsen, 2011) allowed us to conclude that the latter was clearly preferable at the 95% level (i.e. LR=29.57 is greater than 25, the critical χ^2 value for 15 degrees of freedom).

As the ML model can recognise the existence of correlation between the responses of each individual, we expected it to be more informative and also to capture taste variations (detected only systematically in model MNL2) more exhaustively. As with the MNL formulations we began by estimating an ML model similar to the base (MNL1) version but using an error component to allow for correct treatment of the repeated observations by each individual; then, we estimated a second model equivalent to MNL2 with the same approach. The two specifications denoted ECL1 and ECL2 were intended to analyse if incorporating an error component to allow for a correct treatment of the potential correlation between individual responses would improve upon the former MNL estimates. The best estimated models with these specifications are shown in Table 2.

These models are vastly superior to their MNL counterparts (the LR test allows rejecting the null hypothesis that the restricted versions, MNL1 and MNL2, are equivalent to their more general versions, ECL1 and ECL2, with ease); also, ECL2 is superior to ECL1 (LR=30.08 is greater than 26.3, the critical χ^2 value at the 95% level for 16 degrees of freedom). Note also that the ECL parameters are larger than their equivalents in the corresponding MNL models, in line with theoretical expectations (except for the ASC, which must necessarily behave differently) as the scale factor β, not identifiable in Eq. (2), should be greater for each attribute in the ECLs, as already demonstrated by Sillano and Ortúzar (2005).

Table 2
Error components logit models.

Attribute	ECL1		ECL2	
	Parameter value	*t*-test	Parameter value	*t*-test
Metro–Metro	0.43700	3.92	0.24200	2.97
Bus–Metro	0.09310	0.87*	0.06230	0.80*
Metro–Bus	0.18500	1.69*	0.12800	1.61*
Cost	−0.00157	−3.63	−0.00078	−1.38*
Cost–Age			−0.00192	−2.24
Initial Walk	−0.05340	−3.29	−0.04670	−2.01
IWalk-CarLic			−0.01540	−0.48*
Initial Wait	−0.03620	−2.20	−0.02190	−0.92*
IWait-CarLic			−0.02350	−0.74*
Initial Trip	−0.04300	−4.21	−0.04450	−4.29
Trans Walk	−0.03660	−2.28	−0.01480	−0.60*
TWalk-Gender			−0.04510	−1.39*
Trans Wait	−0.07080	4.22	−0.07420	−4.35
Final Trip	−0.04560	−4.36	−0.04690	−4.39
Final Walk	−0.06340	−3.77	−0.08660	−1.95*
FWalk-Gender			0.07530	2.27
Intermodal	−0.14100	−2.45	−0.71600	−3.44
Intermodal-Gender			0.08800	0.76*
Intermodal-Ocup			0.14700	1.09*
Intermodal-WageRate			0.00446	2.50
Intermodal-WalkProp			0.50700	0.83*
Escalator	0.10900	1.99	0.02710	0.18*
Escalator-Gender			0.14100	1.23*
Escalator-N° Trans			0.01290	0.14*
First Avail	0.15100	2.54	0.04090	0.26*
First Avail-CarLic			0.16200	1.43*
First Avail-WaitProp			0.32700	0.54*
Information	0.16300	2.81	0.11000	1.44*
Information-Age			0.15200	1.30*
ε_q	−0.55500	−3.26	0.57100	3.15
Log likelihood	−1132.70		−1117.66	
N° parameters	16		32	

(*) Not significantly different from zero at 95% level

Once the ECL models were analysed we proceeded to estimate a model that was equivalent to MNL1 but using the random parameters logit (RPL) approach to model variations in individual tastes for the attributes in the utility function (while still treating correctly the repeated observations issue, i.e. the panel effect). In a first version, RPL1,we allowed each model parameter in MNL1—except for the ASC—to be distributed over the population; however, we found that the only truly random parameters were those for the *Cost*, *Intermodal Station* and *First Available* attributes (the rest received non-significant variances), so we re-estimated the model, as RPL2, considering the remaining attributes as fixed (A LR test confirmed that this model was not significantly different from the former, so we considered it superior on the parsimony principle.). These model results are shown in Table 3.

3.3. Analysis of model estimation results

In all models the parameter values have signs and values generally consistent with the *a priori* assumptions of the survey. The sole exception was the bus–Metro constant, for which larger values were expected but its estimated values turned out to be small and not significant. This result alters the hierarchy of transfer types initially assumed on the basis of the focus group findings, which suggested that the preference ordering would be Metro–Metro, bus–Metro, Metro–bus and bus–bus (Navarrete, 2010). Nevertheless, since the design included the bus–Metro constant in

Table 3
Random parameters ML models.

Attribute	RPL1		RPL2	
	Parameter value	Standard deviation	Parameter value	Standard deviation
Metro-Metro	0.48700 (4.80)		0.46300 (4.80)	
Bus-Metro	0.10600 (1.13)		0.09710 (1.07)	
Metro-Bus	0.20700 (2.15)		0.20300 (2.16)	
Cost	−0.00162 (−3.19)	0.00377 (4.60)	−0.00158 (−3.22)	0.00352 (4.48)
Initial Walk	−0.05410 (−3.26)	−0.02470 (−0.39*)	−0.05160 (−3.26)	
Initial Wait	−0.03480 (−2.10)	−0.02930 (−0.45*)	−0.03460 (−2.15)	
Initial Trip	−0.04100 (−3.96)	0.00120 (0.04*)	−0.03940 (−3.97)	
Trans Walk	−0.03530 (−2.08)	0.06960 (1.65*)	−0.03560 (−2.15)	0.05830 (1.23*)
Trans Wait	−0.06660 (−3.93)	−0.01020 (−0.24*)	−0.06440 (−3.96)	
Final Trip	−0.04680 (−4.37)	0.00331 (0.14*)	−0.04460 (−4.36)	
Final Walk	−0.06140 (−3.61)	−0.00454 (−0.10*)	−0.06040 (−3.68)	
Intermodal	−0.13000 (−2.19)	−0.23300 (1.49*)	−0.13100 (−2.23)	0.13700 (1.53*)
Escalator	0.13300 (2.30)	−0.03850 (0.21*)	0.12600 (2.25)	
First Avail	0.14800 (2.28)	−0.40700 (3.58)	0.14700 (2.29)	0.39900 (3.55)
Information	0.13700 (2.29)	0.24800 (1.76*)	0.13400 (2.4)	
Log likelihood	−1134.53		−1135.37	
N° Parameters	27		19	

(*) Not significantly different from zero at 95% level

Table 4
Analysis of the distributions of parameters for model RPL2.

Attribute	Expected sign	% of distribution	
		< 0	> 0
Cost	(−)	67.2	32.8
Intermodal	(−)	83.0	17.0
Trans Walk	(−)	73.0	27.0
First Avail	(+)	35.7	64.3

the generation of the survey it was retained in the models. In any case, the non-significance of the result may be due to sample size.

Model ECL2, which captures systematic variations in tastes, was found to be equally consistent with the *a priori* assumptions. Since the structure of this model and its interpretation are more complex, we will describe below the influence of its variables in the utility function.

The ML models with random parameters (RPL) were estimated using different parameter distributions, the Normal distribution proving to be the one with best fit to the population (both for means and variances). To corroborate that their parameter estimates were satisfactory and conformed to the usual microeconomic assumptions, we studied the behaviour of each for model RPL2 in detail (the domain of the Normal distribution is all real values). The result of this test, as displayed in Table 4, is that the expected parameter values according to their respective Normal distributions would be contrary to what is desired in about 30% of cases (see the discussion in Sillano and Ortúzar, 2005). This is fairly high and implies that deeper analysis would be called for before choosing RPL2 as the best model from a purely statistical standpoint.

Models RPL1 and RPL2 are not directly comparable with ECL1 and ECL2 as they are not nested versions of the base model (i.e. with random rather than fixed parameters, they have a different specification and follow a different logic). However, a comparison of their log-likelihood values shows clearly that ECL2 appears to be superior. Also, as the models estimated using an ECL specification are far simpler, in that they just add an error component to take into account the panel effect, they have a clear advantage over the RPL models where it is difficult to differentiate the influence of this effect from that of the distribution of the estimated parameters over the population.

Thus, we concluded that the most informative formulation was ECL2. This result is extremely interesting for it demonstrates once again (see Grisolía and Ortúzar, 2010) that a specification with systematic taste variations and which considers appropriately the correlation between individual responses can be superior to a ML model with random parameters. Furthermore, it offers the enormous advantages of having a much clearer interpretation and being simple to use for estimating willingness to pay.

3.4. Discussion of model ECL2 results

3.4.1. Cost

The parameter for this variable depends on the age of the user. Those under 30 years of age would appear to be 3.46 times more sensitive to cost than those aged 30 and over[6]. This may be due to the fact that young people earn proportionally lower incomes and have more recently set up their own household, with all the expenses that this implies.

3.4.2. Walk times

The *Initial Walk Time* (*IAT*) parameter depends on the number of cars and licensed drivers in the user's household, factors captured by the *CarLic* variable. Thus, as *CarLic* rises the absolute value of the parameter also increases, expressing the fact that the disutility of the initial walk time grows with greater effective access to private car transport. This seems logical given that travel by private vehicle is "door to door" and requires no walking time to access.

The *Transfer Walk Time* (*TAT*) parameter depends on gender, with women penalising this factor 4.05 more times than men. This was made evident during the focus group sessions in which women users expressed greater displeasure at having to transfer due to the aggravations of dealing with jostling crowds at stations, especially when wearing high-heeled shoes.

The *Final Walk Time* (*FAT*) parameter also depends on gender. In this case, however, the disutility for men would appear to be much higher than that for women. This may be explained by the fact that men become more impatient at the final stages of a trip because of the pressure to arrive at their destinations on time (as indicated by them during the focus groups and the actual survey).

[6] The parameter for those under 30 is −0.00078–0.00192 = −0.0027 and this value is 3.46 times larger than −0.00078 which is the value for those 30 and over.

The phenomenon was less noticeable in women, who are either more careful to start their journeys early enough or less subject to strict arrival times.

3.4.3. Wait time

As with *IAT*, the *Initial Wait Time* (*IWT*) parameter depends on the *CarLic* variable, with those having access to a private car penalising this variable more than those who do not. It seems logical as private car travel involves no initial wait or walk time.

3.4.4. Information available

The parameter for this variable depends on age. Those under 30 value the availability of information on transferring 2.38 times more than those 30 or over. The reason behind this could be the greater self-reliance and "online" experience of the younger generation who are more adept at searching for information and expressed annoyance at having to ask others (especially since they are often listening to music on portable media players when riding the system).

3.4.5. Escalator available

The parameter for this variable is influenced both by gender and by the number of transfers users must make. Women have a marked preference for the availability of escalators but for both genders the associated utility grows with the number of transfers that users usually make. These attitudes derive partly from the displeasure, women in particular, express at the walking component of transferring (the escalator thus confers utility by reducing the disutility due to *TAT*) and partly from the fact that with multiple transfers, men as well as women attempt to minimise the physical exertion involved in changing levels within metro stations.

3.4.6. First available bus or train

The parameter for this attribute is affected by two interactions: the *CarLic* variable and the wait time proportion in total individual travel time (denoted *WaitProp*). People with greater access to cars place a much higher value on the ability to board the first arriving bus or train than users who are captives of public transport (or unlikely to travel by private car). This may be attributable to the greater frustration felt by those with frequent access to a transport mode that involves no waiting. Also, as the wait time proportion of a trip increases, so does the utility of boarding the first accessible bus or train due to the frustration inevitably caused by lengthy waits.

3.4.7. Intermodal station

This variable is strongly present in the model and its parameter depends on occupation, gender and personal income as well as on the walk time proportion in total travel time (denoted *WalkProp*). Women report less disutility than men for transferring at intermodal stations during morning peak hours. This may stem from the fact that these stations usually have shopping and other services women find particularly convenient given their household responsibilities.

A marked difference between employed and self-employed workers is also evident in the results. The former group penalises transfers at intermodal stationsless heavily, perhaps because they have set working hours and value the opportunity to make use of the services available at these stations while travelling to work. Another observed tendency is that the greater is the walk time proportion in total travel time the lower is the disutility to passengers of using intermodal stations in the peak. This makes sense if we consider that such stations provide a safe, well-lit environment out of the weather, with access to shopping and services as an added benefit.

Table 5
Willingness to pay for attributes in ECL models.

Attribute	Model ECL1			ECL2
	WTP (Ch$)	**Confidence interval**		**WTP** (Ch$)
		Lower limit (Ch$)	**Upper limit** (Ch$)	
Metro–Metro	278	127	647	154
Bus–Metro	59	−83	243	40
Metro–Bus	118	−19	343	81
Initial Walk	34/min	13/min	83/min	35/min
Initial Wait	23/min	3/min	62/min	23/min
Initial Trip	27/min	13/min	62/min	28/min
Trans Walk	23/min	3/min	62/min	25/min
Trans Wait	45/min	22/min	103/min	47/min
Final Trip	29/min	14/min	66/min	30/min
Final Walk	40/min	18/min	94/min	30/min
Intermodal	−90	−246	−19	−146
Escalator	69	1	192	77
First Avail	92	22	244	124
Information	104	30	262	110

3.5. Estimation of willingness to pay

We estimated the willingness to pay (WTP) for the attributes affecting transfers in the ECL formulations. These subjective values provide a quantitative measure of the effects of these attributes on individual users and define a variable hierarchy by order of influence. The WTP results for the various travel time component parameters (*IAT*, *IWT*, *ITT*, *TAT*, *TWT*, *FTT* and *FAT*) represent the amounts individuals are willing to pay for a minute less of each one, while the values for the transfer attributes (*Intermodal Station*, *Information Available*, *Escalator Available* and *First Available*) indicate the willingness to pay for a trip with these characteristics (in the case of *Intermodal Station*, the value represents a compensation paid to the individual).

The WTP values and associated confidence intervals for model ECL1 were calculated using the methodology of Armstrong et al. (2001), set out in Eq. (6) above, given that the model just has a single error component added to a MNL structure with systematic variations and fixed parameters. Most of the values, which are shown in Table 5, are slightly higher than those obtained for the MNL models and with similar confidence intervals (recall that the benchmark reference for the Metro–Metro, bus–Metro and Metro–bus WTP calculations is the bus–bus transfer, and not the no-transfer option[7]). Table 5 also presents average values for model ECL2 for the sake of simplicity. This model allows computing a large variety of WTP according to the systematic taste variations accounted for in the specification.

Looking at the average values for model ECL2, we can see that the willingness to pay for the *Intermodal Station* attribute proved to have the highest negative value. As the focus groups revealed, users tended to avoid these stations during peak hours; this was clearly an unexpected result confirmed by our modelling analysis and may have interesting policy consequences. On the other hand, the WTP for the *First Available* attribute was the most highly positive, followed by *Information Available* and *Escalator Available*.

[7] The no-transfer option was included in the first pilot survey but was eliminated after it was found that the majority of users would not, under any circumstances, consider a trip that involved a transfer if a no-transfer option was available.

Regarding times, the most penalised among wait and travel times were *Transfer Wait Time* and *Final Travel Time*, as was the case with the MNL models (this also agrees with what was initially assumed).

As for walk times, the MNL models did not generate the expected results according to which *Transfer Walk Time* would be the most penalised time component in the hierarchical ordering and *Initial Walk Time* (*IAT*) and *Final Walk Time* (*FAT*) values would be similar. By contrast, in the superior ECL models, *IAT* and *FAT* were relatively close, yet another indication of the greater explanatory power of these specifications. This finding also tends to confirm that the initial assumptions of the discrete choice experiment were correct.

The WTP values for the ASC represent the amount of money users are willing to pay for trips including a Metro–Metro, bus–Metro or Metro–bus transfer as opposed to a bus–bus combination. Thus, these values should not be treated as the subjective value of transfers directly. The two combinations between bus and Metro have a confidence interval with a negative lower bound due to the fact its *t*-test results are not significant at the 95% level; notwithstanding, the mean values obtained are realistic in magnitude and are fairly similar in both model types.

4. Conclusions

This paper investigated the valuations assigned by users of the *Transantiago* public transport system to the physical characteristics and travel times associated with trips to work during the morning peak hour. The analysis comprised a qualitative component involving user focus groups and a quantitative component consisting of astated choice experiment and the estimation of advanced discrete choice models. Four main physical attributes were identified: intermodal station transfers, ability to board the first accessible bus or train, availability of information and availability of an escalator. Only the first attribute was found to be a disutility, a wholly unexpected result originating incommuters' negative perceptions of the large crowds present at rush hours in the few intermodal stations currently available in the city. The willingness to pay for avoiding a transfer at these stations is in average, according to our best model, 146 Ch$ per trip (approximately one-fourth of the fare) and reflects the greater walk and/or wait times required to transfer in very crowded conditions. The existence of shopping and services at these stations unfortunately adds to the negative views of peak hour users.

Being able to board the first accessible bus or train was the second most valued attribute by users, with a willingness-to-pay valuation of 124 Ch$ per trip in average. This high utility perception derives from the frustration felt by users who are unable to board a bus or train at a stop or station because it is already fully loaded. As for the available information variable, willingness to pay was 110 Ch$ per trip in average. This attribute was appreciated mainly by users under the age of 30 and as was apparent in the qualitative stage of the experiment, this result reflects users' ability to travel without the assistance of others, which would not be the case if transit information were not available.

The last physical attribute in the hierarchy was the availability of an escalator. Though it weighed the least among the four variables considered, it was the one most often mentioned by users, particularly in situations where transfers involved one or more major station-level changes. The willingness to pay for the ability to take an escalator was 77 Ch$ per trip in average. This relatively lower valuation is perhaps due to the fact that users have come to take escalators almost for granted, at least when changing platforms within Metro stations, with the resultant reluctance to pay for it.

The second objective of our experiment was to determine users' relative preferences for the different possible mode transfer combinations. It was assumed *a priori*, based on the notion that users would maximise the perception of speed/frequency and mode reliability, that their preference ordering would be Metro–Metro, bus–Metro, Metro–bus and bus–bus. However, upon conducting the final survey and estimating the models it became obvious that this hierarchy did not totally coincide with the model results. The Metro–Metro and bus–bus transfers were, as predicted, the first and last preferences in this sense; but the utility assigned by users to Metro–bus was almost double than that of bus–Metro. We had originally assumed that transferring from a mode with more uncertainty to one with less uncertainty would be preferred, but this was not reflected in the models (we have not searched for an explanation, but since Metro is considered a superior mode, using it as the main component should be preferable and, typically, in Metro–bus combinations the bus leg is shorter, but this is not necessarily the case otherwise).

The WTP values for the ASC allowed us to determine how much users were willing to pay for a Metro–Metro, bus–Metro or Metro–bus trip as compared to a bus–bus journey. In this case we had to analyse the negativity of the lower bounds for the WTP values of transfers combining bus and Metro, which may have been due to the high variability of the survey responses and led to its low *t*-test values. Nevertheless, the means preserve the hierarchy generated by the models and are realistically sized, with values for the MNL and ML models that are quite similar.

As a caveat, we must note again, that our experiment discarded the "no transfer" option. It was included in the initial design but had to be abandoned after we found in the first pilot survey that most users would simply not consider alternatives involving transfers if the trip could be made directly. Thus, the WTP values we have calculated exclude the "no transfer" alternative. If we were to categorise a direct trip, however, our observations during the survey suggest that its WTP would surely be above that of the Metro–Metro option.

Our final objective was to determine the relationships between the walk, wait and travel time valuations in order to discover which ones penalise most the perception of total trip time. Seven time components were defined for a basic trip with transfers: initial walk, initial wait, first mode travel, transfer walk, transfer wait, second mode travel and final walk. Our model estimates indicate that the most penalised time was the transfer wait time, the WTP value of which was Ch$ 47 per min saved in average according to our best specification. This figure suggests that one of the main targets for future work on improving user perceptions of transferring would be to reduce the associated wait times. A saving of one and a half minutes in this time component is equivalent to the availability of a station escalator and thus represents a major potential resource saving.

The next largest time component disutilities were those associated with the initial and final walk times, whose respective WTP values were 35 and 30 Ch$/min in average. According to data gathered during the focus groups and while conducting the survey, both components were affected by the advent of *Transantiago*. The creation of specific bus stop locations for the new system triggered a change in user perceptions both of accessing and exiting the transit system. Under the old setup, bus drivers were willing to stop wherever passengers desired to board or disembark. Optimising the locations of the new stops could improve current perceptions of these time components.

Acknowledgements

The authors wish to thank John M. Rose, representing Choice Metrics, for his collaboration in providing us with the Ngene software used to build the efficient survey designs. Thanks are also due to Sebastián Raveau, Luis I. Rizzi and María Francisca Yañez for their constant support. The in-depth analysis and useful criticisms of a previous draft of the paper by two anonymous reviewers helped us to improve it considerably. Finally, we are grateful for the partial funding received from the Chilean Fund for Scientific and Technological Research FONDECYT (Project 1050672), the Millennium Institute on Complex Engineering Systems (ICM: P05-004F; FONDECYT: FB016) and ALC-BRT Centre of Excellence.

Appendix A

Table A1.

Table A1
MNL models.

Attribute	MNL1		MNL2	
	Parameter value	*t*-test	Parameter value	*t*-test
Metro–Metro	0.42600	4.72	0.42000	4.61
Bus–Metro	0.08350	0.98*	0.08710	1.01*
Metro–Bus	0.19500	2.20	0.20700	2.30
Cost	−0.00142	−3.59	−0.00071	−1.37*
Cost–Age	–	–	−0.00164	−2.09
Initial Walk	−0.04830	−3.27	−0.04140	−1.97
IWalk-CarLic	–	–	−0.01290	−0.45*
Initial Wait	−0.03240	−2.18	−0.02050	−0.95*
IWait–CarLic	–	–	−0.01920	−0.66*
Initial Trip	−0.03720	−4.00	−0.03860	−4.11
Trans Walk	−0.03290	−2.24	−0.00968	−0.43*
TWalk–Gender	–	–	−0.04470	−1.51*
Trans Wait	−0.06220	−4.08	−0.06420	−4.18
Final Trip	−0.04110	−4.33	−0.04120	−4.30
Final Walk	−0.05790	−3.80	−0.09660	−4.26
FWalk–Gender	–	–	0.06510	2.17
Intermodal	−0.11800	−2.28	−0.58200	−3.11
Intermodal–Gender	–	–	0.05910	0.57*
Intermodal–Ocup	–	–	0.08460	0.69*
Intermodal–WageRate	–	–	0.00413	2.56
Intermodal–WalkProp	–	–	0.53400	0.97*
Escalator	0.11000	2.11	0.00364	0.03*
Escalator-Gender	–	–	0.12800	1.23*
Escalator–N° Trans	–	–	0.02530	0.31*
First Avail	0.13700	2.56	0.00861	0.06*
First Avail–CarLic			0.13200	1.30*
First Avail–WaitProp			0.32100	0.60*
Information	0.13800	2.66	0.07970	1.18*
Information–Age			0.15300	1.46*
Log likelihood	−1141.10		−1126.31	
N° parameters	15		29	

(*) Not significantly different from zero at the 95% level

References

Armstrong, P.M., Garrido, R.A., Ortúzar, J. de D., 2001. Confidence intervals to bound the value of time. Transportation Research 37E, 143–161.
Bierlaire, M., 2003. BIOGEME: a free package for the estimation of discrete choice models. in: Proceedings of the 3rd Swiss Transportation Research Conference, Ascona, Switzerland.
Caussade, S., Ortúzar, J. de D., Rizzi, L.I., Hensher, D.A., 2005. Assessing the influence of design dimensions on stated choice experiment estimates. Transportation Research 39B, 621–640.
Crockett, C., 2002. A Process for Improving Transit Service Connectivity. M.Sc. thesis, Transportation Department, Massachusetts Institute of Technology.
CTPS, 1997. Transfer Penalties in Urban Mode Choice Modelling. Report prepared by the Central Transportation Planning Staff for the Travel Model Improvement Program, FTS, U.S. Department of Transport, Washington, D.C.
Domencich, T., McFadden, D., 1975. Urban Travel Demand: A Behavioural Analysis. North-Holland, Amsterdam.
Espino, R., Martín, J.C., Román, C., 2007. Análisis de la calidad del transporte de pasajeros: un estudio de la disposición a pagar por no transbordar. X Encuentro de Economía Aplicada, Logroño, Spain.
Godoy, G., Ortúzar, J.de D., 2008. On the estimation of mixed logit models. In: Inweldi, P.O. (Ed.), Transportation Research Trends. Nova Science Publishers, Nueva York.
Grisolía, J.M., Ortúzar, J. de D., 2010. Forecasting vs. observed out turn: studying choice in faster inter-island connections. Transportation Research 44A, 159–168.
Guo, Z., 2003. Assessment of the Transfers Penalty for Transit Trips in Downtown Boston. A GIS-Based Disaggregate Modelling Approach. Master in City Planning Dissertation, Transportation Department, Massachusetts Institute of Technology.
Guo, Z., 2008. Transfers and Path Choice in Urban Public Transport Systems. Ph.D. thesis, Transportation Department, Massachusetts Institute of Technology.
Iragüen, P., Ortúzar, J. de D., 2004. Willingness-to-pay for reducing fatal accidents risk in urban areas: an internet-based web page stated preference survey. Accident Analysis and Prevention 36, 513–524.
Jara-Díaz, S.R., 2007. Transport Economic Theory. Elsevier Science, Amsterdam.
Jara-Díaz, S.R., Ortúzar, J. de D., 1989. Introducing the expenditure rate in the estimation of mode choice models. Journal of Transport Economics and Policy 23, 293–308.
Liu, R., Pendyala, R., Polzin, S., 1997. Assessment of intermodal transfer penalties using stated preference data. Transportation Research Record 1607, 74–80.
Louviere, J.J., Hensher, D.A., Swait, J.D., 2000. Stated Choice Methods: Analysis and Applications. Cambridge University Press, Cambridge.
Muñoz, J.C., Ortúzar, J. de D., Gschwender, A., 2009. Transantiago: the fall and rise of a radical public transport intervention. In: Saleh, W., Sammer, G. (Eds.), Travel Demand Management and Road User Pricing: Success, Failure and Feasibility. Ashgate, Farnham.
Navarrete, F.J., 2010. Valoración Subjetiva del Transbordo en Transporte Público: Caso Santiago de Chile. M.Sc. thesis, Department of Transport Engineering and Logistics, Pontificia Universidad Católica de Chile, (in Spanish).
Ortúzar, J. de D., Hutt, G.A., 1984. La influencia de elementos subjetivos en funciones desagregadas de elección discreta. Ingeniería de Sistemas IV, 37–54, in Spanish.
Ortúzar, J. de D., Roncagliolo, D.A., Velarde, U.C., 2000. Interactions and independence in stated preference modelling. In: Ortúzar, J. de D. (Ed.), Stated Preference Modelling Techniques. Perspectives 4. PTRC, London.
Ortúzar, J. de D., Willumsen, L.G., 2011. Modelling Transport, 4th edition John Wiley & Sons, Chichester.
Rose, J.M., Bliemer, M.C.J., 2009. Constructing efficient stated choice experimental designs. Transport Reviews 29, 587–617.
Rose, J.M., Collins A.T., Bliemer M.C.J. and Hensher D.A., 2009a. Ngene, The Cutting Edge in Experimental Design for Stated Choice Experiments. User Manual and Reference Guide. 〈http://choice-metrics.com〉.
Rose, J.M., Hensher, D.A., Caussade, S., Ortúzar, J. de D., Jou, R.-C., 2009b. Identifying differences in willingness to pay due to dimensionality in stated choice experiments: a cross country analysis. Journal of Transport Geography **17**, 21–29.
Sillano, M., Ortúzar, J. de D., 2005. Willingness-to-pay estimation with mixed logit models: some new evidence. Environment and Planning 37A, 525–550.
Torres, I., Greene, M. Ortúzar, J. de D. Valuation of housing and neighbourhood attributes for city centre location: a case study in Santiago. Habitat International, 39, http://dx.doi.org/10.1016/j.habitatint.2012.10.007, in press.
Train, K., 2009. Discrete Choice Methods with Simulation. 2nd edition Cambridge University Press, Cambridge.
Wardman, M., Hine, J., Stradling, S., 2001. Interchange and Travel Choice, 1. Scottish Executive Central Research Unit, Glasgow.